Though the theoretical value of velocity ratio is calculated as above, it differs from it because of the thickness of the belt and belt slip. These factors should also be taken into account in calculating the actual velocity ratio.

The speed of the shaft or the pulley is expressed in RPM (Revolutions Per Minute). If we want to increase the speed of the follower with respect to the driver, the pulley on the driven shaft should be smaller in size (diameter) than the pulley on the driving shaft. If we want to decrease the speed of the follower, the pulley on the driven shaft should be larger in size.

$$D_1N_1 = D_2 N_2$$

EXAMPLE 1

Pulley of diameters 360 mm and 60 mm are connected by a belt drive. Find the velocity ratio.

D_1 = 360 mm

D_2 = 60 mm

Driver pulley speed = N_1

Driven pulley speed = N_2

Driving pulley diameter(D_1) = 360 mm

Driven pulley diameter (D_2) = 60 mm

$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{D_1}{D_2} = \frac{6}{1} = \frac{360}{60} = 6{:}1$$

Velocity ratio = 6:1

EXAMPLES 2

Two pulleys of diameters of 500 mm and 250mm are connected by means of a open belt drive. If the larger pulley rotates at a speed of 400 rpm in clockwise direction, find the speed and direction of rotation of the smaller pulley.

D_1 = 500 mm D_2 = 250 mm

N_1 = 400 rpm

Driving pulley diameter (D_1) = 500 mm

Driven pulley diameter (D_2) = 250 mm

Driver pulley speed N_1 = 400 rpm

Driven pulley speed N_2 = ?

$$D_1N_1 = D_2 N_2$$

$$N_2 = \frac{D_1N_1}{D_2}$$

$$= \frac{500 \times 400}{250}$$

N_2 = 800 rpm clockwise direction

EXAMPLE 3

Two shafts are connected by a belt drive. On one of the shafts, a pulley of 200 mm diameter is fitted and it rotates at a speed of 3000 rpm in anticlockwise direction. What should be the diameter of the driven pulley if it is to rotate at a speed of 1500 rpm in clockwise direction? What should be the type of belt driver?

D_1 = 200 mm D_2 = ?

N_1 = 3000 rpm N_2 = 1500 rpm

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